

Project Update Report

This report describes my individual progress in the Logic Puzzle Solver project for the CSE 440 Artificial Intelligence course. The research applies Constraint Satisfaction Problem (CSP) approaches to solve Sudoku, and my task is constraint propagation.

For the first three weeks I worked on learning the concepts under constraint-propagation, developing a pseudocode for forward checking and AC-3, and implementing the first draft of the Python framework for forward checking

Week 1 (24th June – 30th June): Studying Constraint Propagation

- **CSP Representation:** I learned how Sudoku is represented as a constraint achievement problem, where each cell is a variable and its possible numbers are the domain.
- **Consistency Concepts:** In order to understand how incorrect values are eliminated, I covered node consistency, arc consistency, domain reduction, and constraint checking.
- **Forward Checking:** I learned how an assignment eliminates the same value from the domains of unassigned cells in the relevant column, row, and 3X3 box.
- **Practical Verification:** I used simplified Sudoku examples and considered an assignment invalid if the domain of a neighboring cell became empty.

The work presented a clear conceptual basis for implementing constraint propagation

Week 2 (1st July – 7th July): Preparing Algorithms and Pseudocode

- **Algorithm Planning:** I drafted pseudocode for AC-3 and forward checking algorithms in an organized way.
- **AC-3 Queue Setup:** I set up the initial queue of connected Sudoku cells such that one arc could be checked at a time.
- **Domain Revision:** I planned the steps for revising domains, eliminating inconsistent values, and adding affected neighboring arcs back to the queue.

- **Failure Detection:** I implemented a failure condition that halts the process if the domain of any Sudoku cell is empty.
- **Forward Checking Pseudocode:** I described how to remove an assigned value from related unassigned cells and how to reject an invalid assignment right away.

This provides an organized framework for implementing and evaluating the constraint-propagation approaches.

Week 3 (8th Jul – 14th July): Developing the Forward Checking Framework

- **Module Development:** I started developing a Python module for forward checking.
- **Input Function:** I developed a function that takes a chosen Sudoku cell and its assigned value as input.
- **Neighbor Identification:** I implemented logic to detect all related cells in the same row, column, and 3×3 box.
- **Domain Updating and Validation:** Removed assigned value from neighboring domains, verified for empty domains, and returned success or failure.
- **Change Recording and Restoration:** Record removed domain values and support restoring them when the solver backtracks.

In order to enhance the speed and effectiveness of the problem solver, I will incorporate AC-3, full forward checking, and test both approaches on various Sudoku puzzles.